EPSR-test-ex93 Gaussian basis of the empirical potential (Gaussian T)

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the Gaussian representation of the empirical potential (EP) in the `epsr` engine of EPSR [1,2], selected with `Gaussian` in the `.pcof` file instead of the default Poisson (power-exponential) functions [3]. The basis determines how the difference between data and simulation is expanded in Q and converted into an r-space potential.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `inita` (`T` -> `gaussian=.true.`, value rewritten as `Gaussian`)
- `epsr_standalone_rev3.f`: `ptgnsq` with `calc_gaussq` (Q-space basis), `fit_expansion`
- `epsr_standalone_rev3.f`: `generate_ep` with `calc_gaussr_grad` (r-space potential and gradients)

Rust (port, rust/epsr-core/src/):

- `structure.rs`: `inita`
- `potential.rs`: `ptgnsq`, `calc_gaussq`, `fit_expansion`
- `potential.rs`: `generate_ep`, `calc_gaussr_grad`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. For this case the EP is a sum of Gaussians centred at d = m·pdstep with width `psigma2`; in Q space each basis function is sin(Qd)/(Qd)·exp(-(Q·sigma)^2/2).

Inputs

Made from EPSR-test with `scripts/make_case.py`: `.pcof` keyword `Gaussian T` instead of `Poisson`. The EP coefficients fitted with the Poisson basis in EPSR-test are kept and reinterpreted in the Gaussian basis.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
The value `T` is rewritten as `Gaussian` after pass 1, so passes 2 and 3 test the second accepted spelling. The basis is active: the README reports an EP energy of about 99 compared with 3.7 in the Poisson run. The README reports a gcov check of the Fortran covering `inita`, `generate_ep`/`calc_gaussr_grad` and `ptgnsq`/`calc_gaussq`.

Notes

`rbroad`, `dmax` and `rho` are not used by the Gaussian basis routines, and with `rmin` = `0` `ncfirst` is never touched.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.